

# Physical Chemistry (II)

## Lecture 18

### Electronic Structure Calculation



# Treating Helium

- Perturbation and variational approaches
  - can give very accurate energies
  - computationally difficult
  - not transferable to large atoms and molecules
- Orbital approximation
  - Each electron's spatial part is represented by an one-electron orbital  $\psi_i(\vec{r})$
  - Common concept in chemistry
  - How to obtain “accurate”  $\psi_i(\vec{r})$ ?



# Hartree-Fock Method

- The **self-consistent field** (SCF) method
- The atomic wavefunction = **Slater determinant**

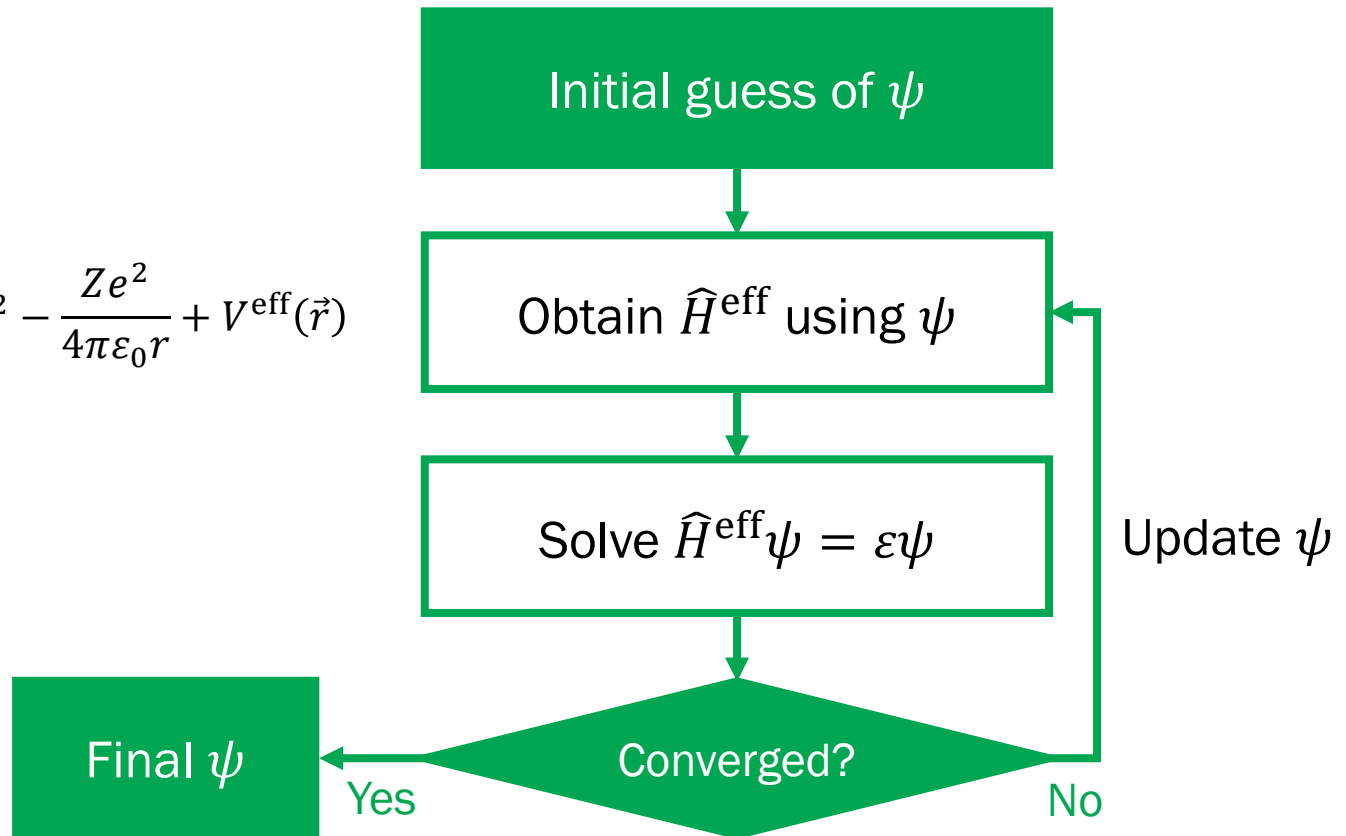
For  $N$  quantum states  $a, b, \dots, n$ ,

$$\phi_{ab\dots n}(1, 2, \dots, N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} u_a(1) & u_b(1) & \cdots & u_n(1) \\ u_a(2) & u_b(2) & \cdots & u_n(2) \\ \vdots & \vdots & \ddots & \vdots \\ u_a(N) & u_b(N) & \cdots & u_n(N) \end{vmatrix}$$

Last class

# SCF Method

$$\hat{H}^{\text{eff}} = -\frac{\hbar^2}{2m_e} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} + V^{\text{eff}}(\vec{r})$$



Last class

# Ground-State Energy

$$E = \sum_{j=1}^M 2E_j + \sum_{i=1}^M \sum_{j=1}^M (2J_{ij} - K_{ij})$$

$$\varepsilon_i = E_i + \sum_{j=1}^M (2J_{ij} - K_{ij})$$

$$\therefore E = \sum_{i=1}^M (E_i + \varepsilon_i)$$

# Koopman's Approximation

$$IE \approx -\varepsilon_i$$

- Ionization energy gives information about orbital energies
- Good approximation for closed-shell systems
- Photoelectron spectroscopy: experimental method to measure ionization energy

# Functional Form of Orbitals

- $N$ -electron Hamiltonian: depends only on **distance**

$$\hat{H} = \sum_{j=1}^N \left( -\frac{\hbar^2}{2m_e} \nabla_j^2 - \frac{Ze^2}{4\pi\epsilon_0 r_j} \right) + \sum_{j=1}^N \sum_{i < j} \frac{e^2}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|}$$

- **Angular part** of orbitals: H-atom wavefunctions
  - Spherical harmonics  $Y_l^{m_l}(\theta, \phi)$
  - Realized form  $\frac{i^{(1 \mp 1)/2}}{\sqrt{2}} \{Y_l^{m_l}(\theta, \phi) \pm [Y_l^{m_l}(\theta, \phi)]^*\}$

# Radial Part

- H-atom radial wavefunctions

$$R_{10}(r) = 2 \left( \frac{Z}{a_0} \right)^{3/2} e^{-Zr/a_0}$$

$$R_{20}(r) = \left( \frac{Z}{2a_0} \right)^{3/2} \left( 2 - \frac{Zr}{a_0} \right) e^{-Zr/2a_0}$$

$$R_{21}(r) = \frac{1}{\sqrt{3}} \left( \frac{Z}{2a_0} \right)^{3/2} \frac{Zr}{a_0} e^{-Zr/2a_0}$$

- We can use ...

$$R_n(r) \propto \left( \frac{\zeta r}{a_0} \right)^{n-1} e^{-\zeta r/a_0}$$



# Slater-Type Orbitals (STOs)

$$S_{nlm_l}(r, \theta, \phi; \zeta) = \frac{(2\zeta/a_0)^{3/2}}{\sqrt{(2n)!}} \left( \frac{2\zeta r}{a_0} \right)^{n-1} e^{-\zeta r/a_0} Y_l^{m_l}(\theta, \phi)$$

- $\zeta$ : orbital exponent
- Linear combination of STOs can be used to model orbitals
  - e.g. double-zeta orbital

$$\phi_{1s}(\vec{r}) = c_1 S_{1s}(\vec{r}; \zeta_1) + c_2 S_{1s}(\vec{r}; \zeta_2)$$

## Example: He 1s

For a helium atom, one can use five STOs:

$$\phi_{1s}(\vec{r}) = c_1 S_{1s}(\zeta_1) + c_2 S_{1s}(\zeta_2) + c_3 S_{1s}(\zeta_3) + c_4 S_{1s}(\zeta_4) + c_5 S_{1s}(\zeta_5)$$

By the Hartree-Fock method, one can optimize

$$\begin{aligned}\phi_{1s}(\vec{r}) = & 0.768 S_{1s}(\zeta = 1.417) + 0.223 S_{1s}(\zeta = 2.377) \\ & + 0.00408 S_{1s}(\zeta = 4.396) - 0.00994 S_{1s}(\zeta = 0.527) \\ & + 0.00230 S_{1s}(\zeta = 7.843)\end{aligned}$$

## Example: He 1s

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The optimized energy is

$$E = -2.861680 E_h \text{ (exp. } -2.9033 E_h)$$

- Determination of  $\zeta$  takes most of computational time  
→ We can “preset” the values of  $\zeta$  and optimize

# Preset Orbital Exponents

| Atom | $\zeta_{1s}$ | $\zeta_{2s} = \zeta_{2p}$ |
|------|--------------|---------------------------|
| H    | 1.24         |                           |
| He   | 1.69         |                           |
| Li   | 2.69         | 0.80                      |
| Be   | 3.68         | 1.15                      |
| B    | 4.68         | 1.50                      |
| C    | 5.67         | 1.72                      |
| N    | 6.67         | 1.95                      |
| O    | 7.66         | 2.25                      |
| F    | 8.56         | 2.55                      |
| Ne   | 9.64         | 2.88                      |

# Basis Set

- Atomic orbitals  $\psi$  can be represented by linear combinations of some functions  $\phi_j$

$$\psi(\vec{r}) = \sum_{j=1}^K c_j \phi_j = c_1 \phi_1 + c_2 \phi_2 + \cdots + c_K \phi_K$$

- **Basis functions:**  $\phi_j$  (e.g. STOs)
- **Basis set:**  $\{\phi_j\}$

# Gaussian-Type Orbitals (GTOs)

- In the HF calculation, we must do numerical integration:

$$J_{ij} = \int \psi_i^*(\vec{p})\psi_j^*(\vec{q}) \frac{e^2}{4\pi\epsilon_0|\vec{p} - \vec{q}|} \psi_i(\vec{p})\psi_j(\vec{q}) d\vec{p}d\vec{q}$$

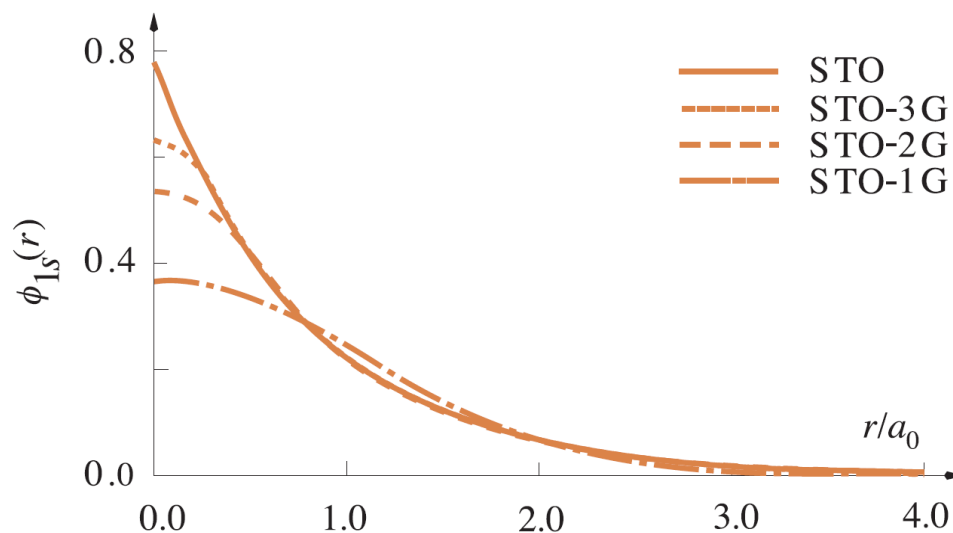
$$K_{ij} = \int \psi_i^*(\vec{p})\psi_j^*(\vec{q}) \frac{e^2}{4\pi\epsilon_0|\vec{p} - \vec{q}|} \psi_i(\vec{q})\psi_j(\vec{p}) d\vec{p}d\vec{q}$$

- Gaussian-type orbitals (GTOs): computationally efficient

$$G_{ijk}(r, \theta, \phi; \alpha) = Nr^{n-1}e^{-\alpha r^2}Y_l^m(\theta, \phi)$$

# STO-NG: Fitting Trick

$$S_{nlm}(r, \theta, \phi) \leftarrow \sum_{i=1}^N d_i G_{nlm}(r, \theta, \phi; \alpha_i)$$



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Fig. 12.3)

# Split-Valence Basis Set

- Inner-shell electrons: a single orbital
- Valence-shell electrons: a sum of orbitals

e.g. Lithium ( $d_i$  and  $\alpha_i$  are preset)

$$\phi_{1s}(\vec{r}) = d_1 G_{1s}(\alpha_1) + d_2 G_{1s}(\alpha_2) + d_3 G_{1s}(\alpha_3)$$

$$\phi_{2s}(\vec{r}) = c_1 \phi_{2s,1}(\vec{r}) + c_2 \phi_{2s,2}(\vec{r})$$

$$\phi_{2s,1}(\vec{r}) = d_4 G_{2s}(\alpha_4) + d_5 G_{2s}(\alpha_5)$$

$$\phi_{2s,2}(\vec{r}) = G_{2s}(\alpha_6)$$



RS 11.2

M 12.3

# ***N-MPG* Basis Sets**

- *N*: number of Gaussian functions for each inner-shell electron orbital
- *M*, *P*: numbers of Gaussian functions for each valence-shell electron orbital
- 3-21G, 6-31G, 6-311G, ...



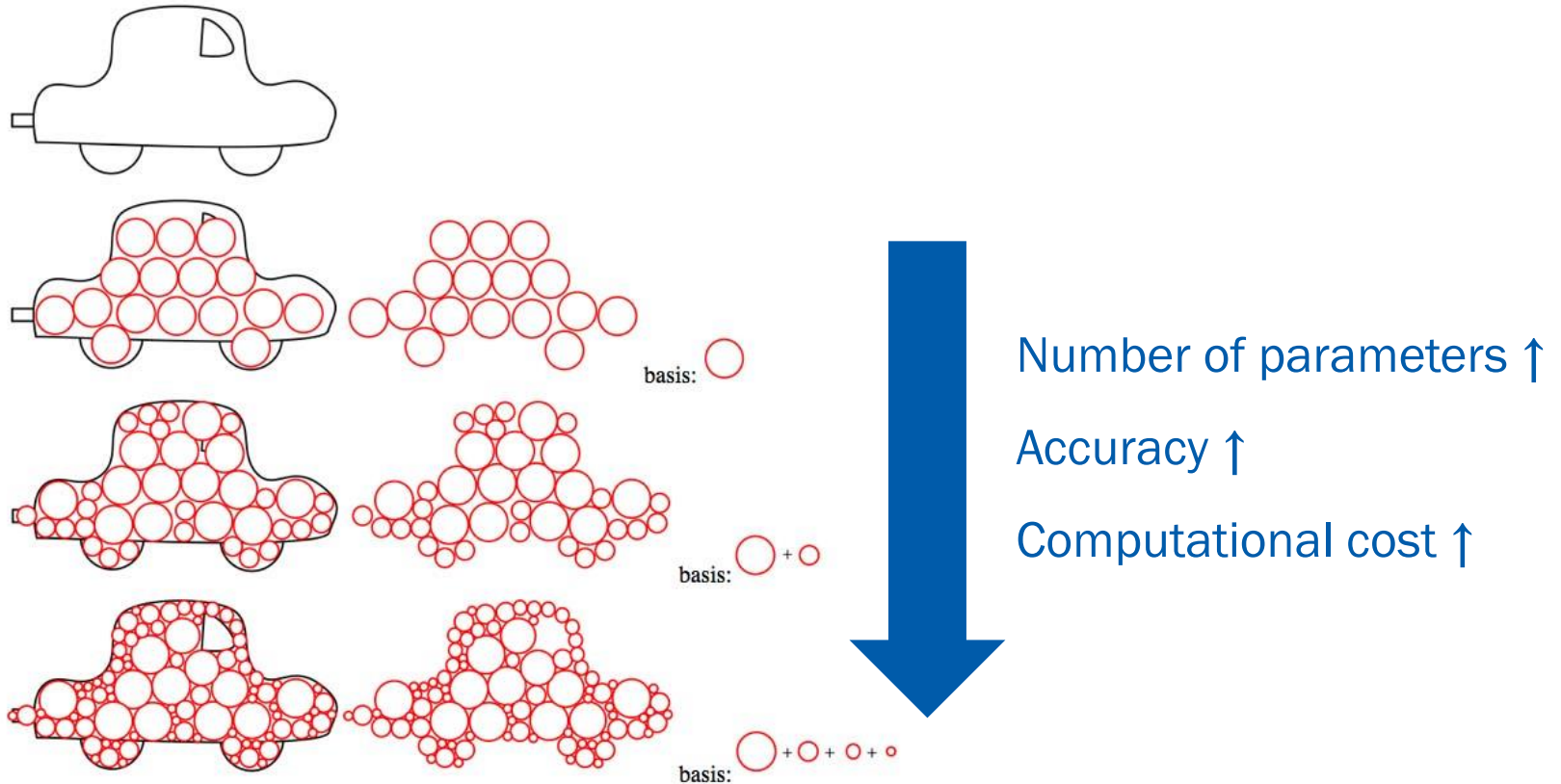
# Polarization and Diffuse Functions

- Polarization functions
  - \*: *d*-type Gaussian functions added to Li-Ca 2*p* orbitals
  - \*\*: *p*-type Gaussian functions added to H and He 1*s* orbitals
  - 6-31G\*, 6-31G\*\*, ...
- Diffuse functions
  - +: Gaussian functions with small exponents on non-H atoms
  - ++: Gaussian functions with small exponents on all atoms
  - 3-21+G, 6-31+G\*, 3-21++G, 6-31++G\*, ...



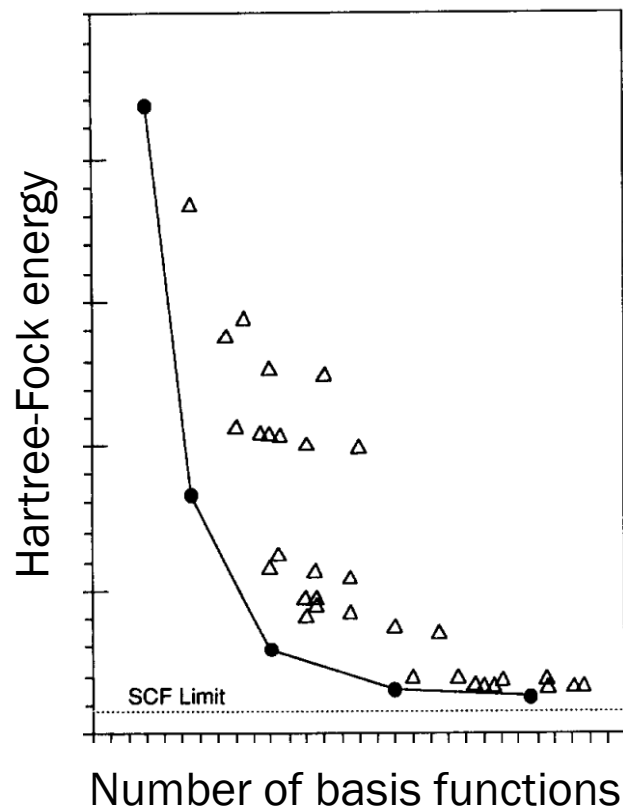
RS 11.2  
M 12.4

# Size of Basis Set



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# Complete Basis Set (CBS) Limit



(Image: Petersson et al., *J. Chem. Phys.*, 1988)

# Hartree-Fock Method

- The self-consistent field (SCF) method
- The atomic wavefunction = Slater determinant
- Electrons interact only through an average potential  
→ They are “uncorrelated”
- Correlation energy  $E_{\text{corr}}$

$$E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}}$$

# Extension of HF

- Start from one-electron wavefunctions (“orbitals”)  $u_i$

$$\phi = |u_a \quad u_b \quad \cdots \quad u_n|$$

- HF determinants use **occupied orbitals only**
- **Inclusion of unoccupied (virtual) orbitals** may increase accuracy and take care of correlation

# Example: Virtual Orbital of He

- Double zeta orbital

$$\phi_{1s}(\vec{r}) = c_1 S_{1s}(\vec{r}; \zeta = 1.454) + c_2 S_{1s}(\vec{r}; \zeta = 2.911)$$

- Ground-state orbital (by Hartree-Fock)

$$\psi_0 = 0.844 S_{1s}(\zeta = 1.454) + 0.181 S_{1s}(\zeta = 2.911)$$

- Virtual orbital (orthogonal to  $\psi_0$ )

$$\psi_v = 1.624 S_{1s}(\zeta = 1.454) - 1.821 S_{1s}(\zeta = 2.911)$$

# Example: Virtual Orbital of He

- Hartree-Fock wavefunction  $\phi_{\text{HF}} = \psi_0(\vec{r}_1)\psi_0(\vec{r}_2)$
- Better wavefunction

$$\phi_{\text{CI}} = c_1\psi_0(\vec{r}_1)\psi_0(\vec{r}_2) + c_2\psi_v(\vec{r}_1)\psi_v(\vec{r}_2)$$

Using a variational method, one can obtain

$$\phi_{\text{CI}} = 0.999 \psi_0(\vec{r}_1)\psi_0(\vec{r}_2) - 0.0476 \psi_v(\vec{r}_1)\psi_v(\vec{r}_2)$$

with optimized energy  $E = -2.87541 E_{\text{h}}$

(HF energy  $E = -2.86167 E_{\text{h}}$ )



RS 11.4 & 11.5  
M 12.7

# Post-HF Methods

- Configuration interaction (CI)
  - Coupled-cluster (CC) theory
  - Møller-Plesset (MP) perturbation theory
- } *ab initio* methods
- Density functional theory (DFT)



Nobel prize in Chemistry 1998

# Configuration Interaction

- Full configuration interaction (full CI)

$$\phi_{\text{CI}} = c_0 \phi_0 + \underbrace{\sum_i \sum_a c_i^a \phi_i^a}_{\text{"singles"}} + \underbrace{\sum_{i,j} \sum_{a,b} c_{ij}^{ab} \phi_{ij}^{ab}}_{\text{"doubles"}} + \dots$$

→ **Exact solution** (in the limit of an infinite basis set)

- Approximations: CI doubles (CID), CI singles and doubles (CISD), ...
  - Note: CI singles do not yield correct physics for ground state

# Coupled-Cluster Theory

For an  $N$ -electron system, the exact wavefunction is

$$\psi_{\text{exact}} = e^{\hat{T}} \phi_0$$

where  $\hat{T} = \hat{T}_1 + \hat{T}_2 + \cdots + \hat{T}_N$  and  $\hat{T}_i$  is an excitation operator.

- CC doubles (CCD)

$$\phi_{\text{CCD}} = e^{\hat{T}_2} \phi_0 = \phi_0 + \hat{T}_2 \phi_0 + \frac{1}{2} \hat{T}_2^2 \phi_0 + \cdots$$

- CC singles and doubles (CCSD), ...

# Møller-Plesset Perturbation Theory

- **Unperturbed wavefunction:** Hartree-Fock wavefunction

$$\hat{H}_i^{(0)} = -\frac{\hbar^2}{2m_e} \nabla_i^2 - \frac{Ze^2}{4\pi\epsilon_0 r_i} + V^{\text{eff}}(\vec{r}_i)$$

- **Perturbation:** difference

$$\hat{H}_i^{(1)} = \sum_j \frac{e^2}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|} - V^{\text{eff}}(\vec{r}_i)$$

- **Notation:** MP $n$  ( $n$ : order)  $\rightarrow$  MP2, MP3, MP4, ...

# Density Functional Theory

- $N$ -particle wavefunction:  $\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)$   
→  $3N$  independent variables
- Electron density

$$\rho(\vec{r}) = N \int \psi^*(\vec{r}, \vec{r}_2, \dots, \vec{r}_N) \psi(\vec{r}, \vec{r}_2, \dots, \vec{r}_N) d\vec{r}_2 \cdots d\vec{r}_N$$

→ 3 independent variables

Use electron density instead of wavefunction!

# Hohenberg-Kohn Theorems

- System energy can be written as a function of  $\rho$ :

$$E = E[\rho]$$

- For any trial electron density  $\rho$ ,

$$E_0 \leq E[\rho]$$

for true ground-state energy  $E_0$  (variational principle).

# Kohn-Sham Equation

- Kohn-Sham orbitals: non-interacting “fictitious” particles

$$\rho(\vec{r}) = \sum_i |\phi_i(\vec{r})|^2$$

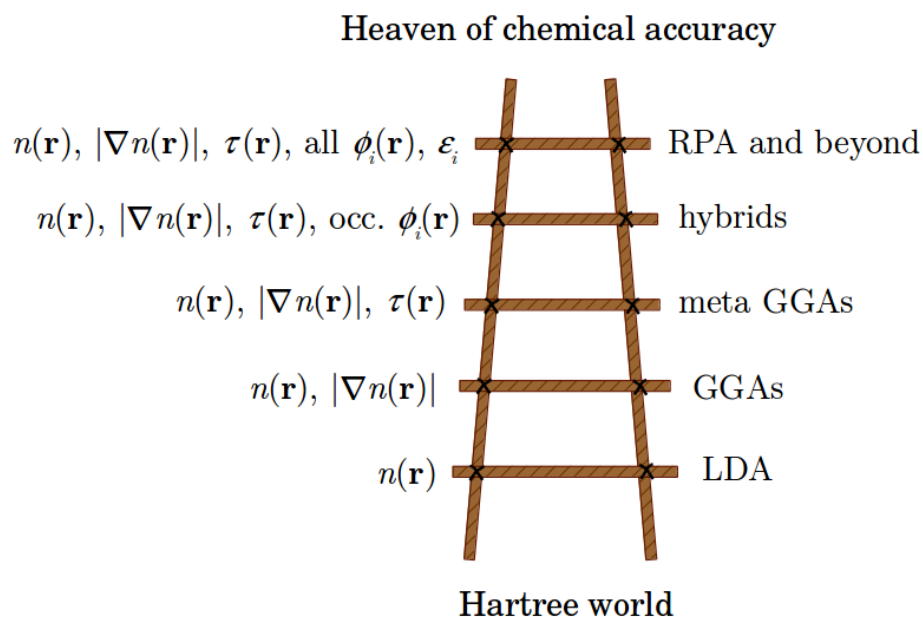
- Kohn-Sham equation

$$\left[ -\frac{\hbar^2}{2m_e} \nabla^2 + V^{\text{eff}}(\vec{r}, \rho(\vec{r})) \right] \phi_i(\vec{r}) = \varepsilon_i \phi_i(\vec{r})$$

$$V^{\text{eff}}(\vec{r}, \rho(\vec{r})) = V_{\text{nuc-e}}(\vec{r}) + e^2 \int \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} d\vec{r}' + \frac{\delta E^{\text{XC}}[\rho(\vec{r})]}{\delta \rho(\vec{r})}$$

# Exchange-Correlation Term

- The exact form of  $E^{\text{XC}}[\rho(\vec{r})]$  is unknown
- Many forms have been suggested (LDA, GGA, hybrid, ...)

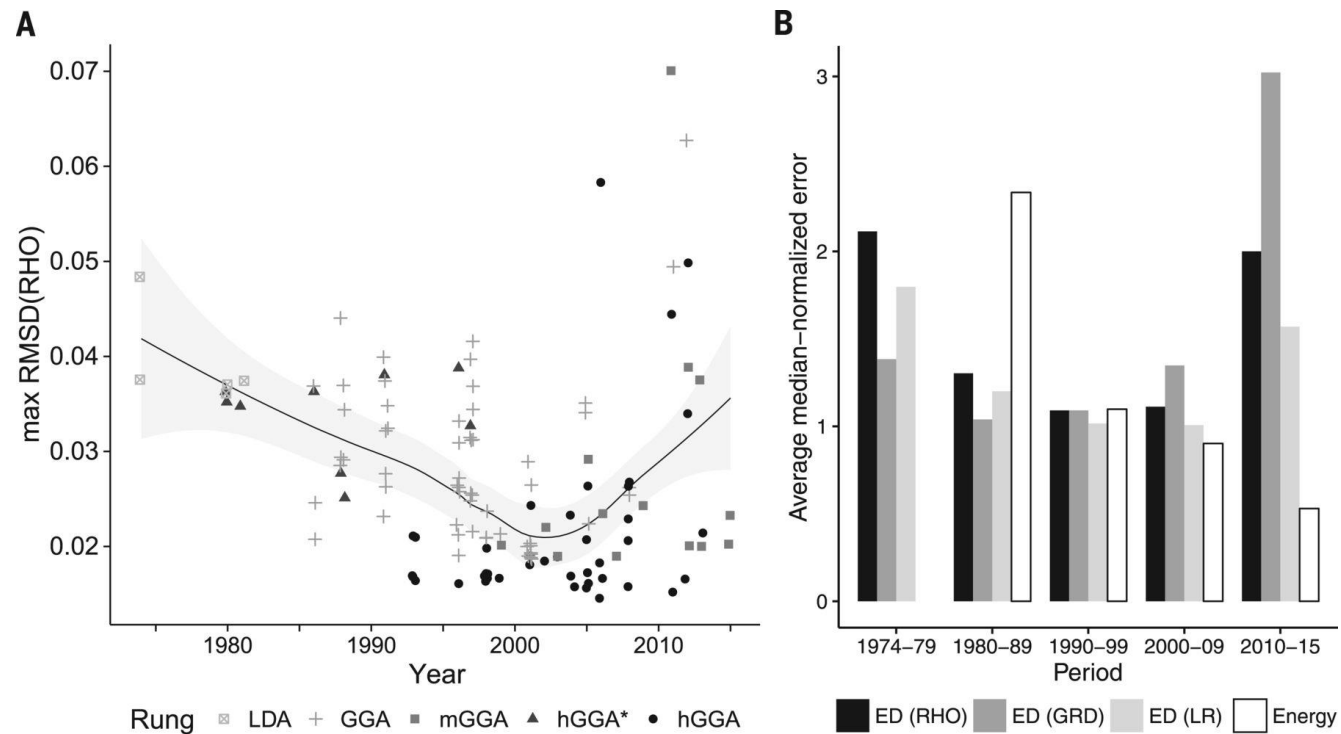


(Image: <https://openscholar.huji.ac.il/elikraisler/density-functional-theory>)



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# DFT: Are We Doing it All Right?



(Image: Medvedev et al., *Science* 2017)